EMG4_preprocessing_data

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Contents

Overview

This document contains the preparation of the EMG4 data for the Multilevel analysis.

```r
library(readxl)
library(sjPlot)
library(tidyverse)
```

check whole trial data

check number of rows per sjCode

```r
# check number of rows per sjCode, should be 64,
# except for 2924 for whom Markertrial 32 is excluded,
# see data_check_preparation_notes
table(whole_trial_data$sjCode)
```

```
##
## 1055 1088 1124 1145 1258 1278 1520 1707 1734 1802 2009 2020 2069 2116 2323 2528
##   64   64   64   64   64   64   64   64   64   64   64   64   64   64   64   64
## 2924 3074 3313 3555 3583 3866 3880 4399 4463 4806 4808 5029 5066 5884 5959 6012
##   63   64   64   64   64   64   64   64   64   64   64   64   64   64   64   64
## 6041 6087 6169 6551 7015 7030 7063 7334 7417 7716 7735 7995 8018 8081 8320 8381
##   64   64   64   64   64   64   64   64   64   64   64   64   64   64   64   64
## 8522 8569 8683 8769 8800 8811 9080 9097 9131 9144 9166 9600
##   64   64   64   64   64   64   64   64   64   64   64   64
```

```r
# check number of rows/trials per Markertrial. should be max 60 for all 51 Markertrials.
# Could also be lower if a trial is excluded, see comment about Markertrial 32.
table(whole_trial_data$Markertrial)
```

```
##
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
## 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60
## 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
## 60 60 60 60 60 59 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60 60
## 53 54 55 56 57 58 59 60 61 62 63 64
## 60 60 60 60 60 60 60 60 60 60 60 60
```

```r
MissingMarkertrials <- whole_trial_data %>%
  select(sjCode, Markertrial) %>%
  group_by(Markertrial) %>%
  summarise(Freq = n()) %>%
  filter(Freq != 60)
table(MissingMarkertrials)
```

```
##            Freq
## Markertrial 59
##          32  1
```

```r
# check number of trials per MoralityCondition, should be max 32*60=1920 per condition
table(whole_trial_data$MoralityCondition)
```

```
##
##  102  103
## 1919 1920
```

```r
# check number of trials per StateAdjectiveCondition, should be max 32*60=1920 per condition
table(whole_trial_data$StateAdjectiveCondition)
```

```
##
##  104  105
## 1919 1920
```

```r
# Baseline should be between 500 and 2000
table(whole_trial_data$Baseline)
```

```
##
##  500  550  600  650  700  750  800  850  900  950 1000 1050 1100 1150 1200 1250
##   61    4   89    5   89    5  112    4  119    4  135    1  150    3  143    5
## 1300 1350 1400 1450 1500 1550 1600 1700 1750 1800 2000
##  161    3  161    1  753   25 1068  685    8   12   33
```

```
summary(whole_trial_data$Baseline)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     500    1300    1500    1413    1600    2000
```

```
# Identify a time bin with extreme outlier
x <- whole_trial_data %>%
  filter(Markertrial == 45 & MoralityCondition== 103 & StateAdjectiveCondition ==105)  %>%
  select(sjCode,CorrPr100ms_288)
mean <- mean(x$CorrPr100ms_288)
sd <- sd(x$CorrPr100ms_288)
outlier <- x %>%
  filter(sjCode == 8800)
(outlier$CorrPr100ms_288-mean)/sd
```

```
## [1] 3.749852
```

```
# Participant 8800 has an extreme outlier (z=3.75) at trial 45 in time bin 288.
# This data point will be removed from the whole trial figures.
```

Load data from all 3 segments

Scatterplot:

Preprocessing

Preprocessing is needed to transform the data from wide to long format based on the 100ms bins for the corrugator response. The relevant variables are computed for the different conditions. Another MoralityValence variable is computed to estimate the final no-intercept model. Furthermore, we've computed linear, quadratic and cubic time trends for each condition (moral, immoral).

1. Morality segment

```
# convert from wide to long format for time and select only corrugator
mor_long <- mor_data_raw %>%
  select(!starts_with("MoralityZygoPr100ms")) %>%
  pivot_longer(
    cols = starts_with("MoralityCorrPr100ms"),
    names_to = "time",
    names_prefix = "MoralityCorrPr100ms_",
    names_transform = list(time = as.integer),
```

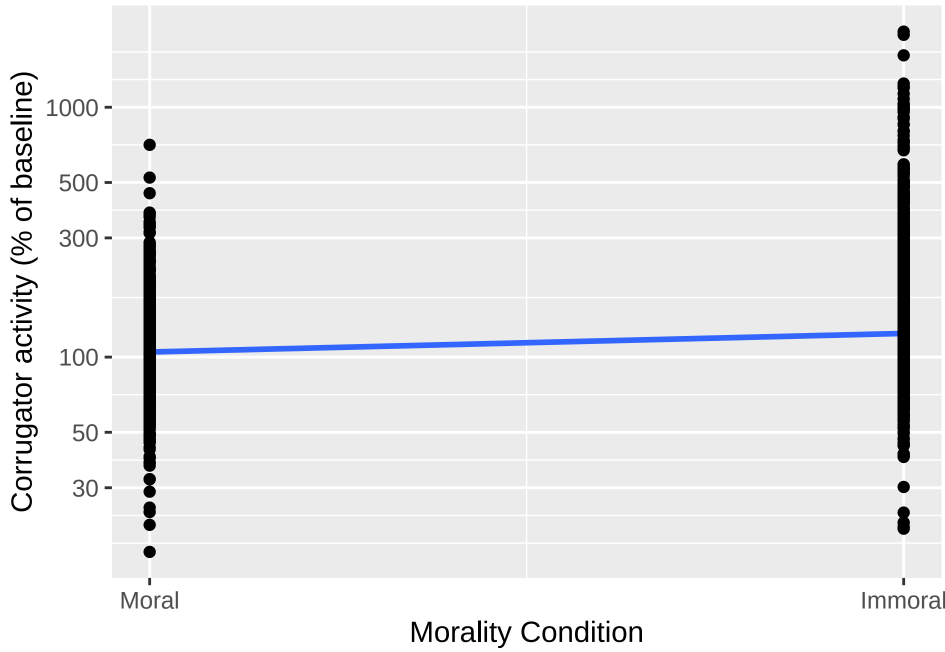

Figure 1: Participants' mean corrugator activation during the morality segment, per Morality Condition

```
    values_to = "Morality_response"
  )
str(mor_long)
```

```
## tibble [191,950 x 13] (S3: tbl_df/tbl/data.frame)
##  $ sjCode                   : num [1:191950] 1055 1055 1055 1055 1055 ...
##  $ SjNr                     : num [1:191950] 1 1 1 1 1 1 1 1 1 1 ...
##  $ Trial                    : num [1:191950] 1 1 1 1 1 1 1 1 1 1 ...
##  $ incCorr                  : num [1:191950] 1 1 1 1 1 1 1 1 1 1 ...
##  $ incZygo                  : num [1:191950] 1 1 1 1 1 1 1 1 1 1 ...
##  $ Markertrial              : num [1:191950] 52 52 52 52 52 52 52 52 52 52 ...
##  $ MoralityCondition        : num [1:191950] 102 102 102 102 102 102 102 102 102 102 ...
##  $ StateAdjectiveCondition: num [1:191950] 104 104 104 104 104 104 104 104 104 104 ...
##  $ Baseline                 : num [1:191950] 600 600 600 600 600 600 600 600 600 600 ...
##  $ baseCorr                 : num [1:191950] 3.97 3.97 3.97 3.97 3.97 ...
##  $ baseZygo                 : num [1:191950] 6.4 6.4 6.4 6.4 6.4 ...
##  $ time                     : int [1:191950] 1 2 3 4 5 6 7 8 9 10 ...
##  $ Morality_response        : num [1:191950] 80.2 100.1 112.6 93.9 105.2 ...
```

```
# moral(102)* pos(104), neg(105) = 608, 710
# immoral(103)* pos(104), neg(105) = 712, 815

# compute relevant factors for the analysis
mor_long <- mor_long %>%
  mutate(MoralityValence = MoralityCondition*StateAdjectiveCondition-10000)

#compute linear, quadratic and cubic time in order to compute time trends per condition
linear_time_centered <- (mor_long$time-25.5)/10
```

```r
unique(linear_time_centered)
```

```
##  [1] -2.45 -2.35 -2.25 -2.15 -2.05 -1.95 -1.85 -1.75 -1.65 -1.55 -1.45 -1.35
## [13] -1.25 -1.15 -1.05 -0.95 -0.85 -0.75 -0.65 -0.55 -0.45 -0.35 -0.25 -0.15
## [25] -0.05  0.05  0.15  0.25  0.35  0.45  0.55  0.65  0.75  0.85  0.95  1.05
## [37]  1.15  1.25  1.35  1.45  1.55  1.65  1.75  1.85  1.95  2.05  2.15  2.25
## [49]  2.35  2.45
```

```r
quadratic_time_centered <- linear_time_centered*linear_time_centered
cubic_time_centered <- linear_time_centered*quadratic_time_centered

#compute time trends per condition and filter only the included trials based on incCorr and incZygo
mor_long_inc <- mor_long %>%
  mutate(t1_moral = ifelse(MoralityCondition == 102, linear_time_centered, 0)) %>%
  mutate(t2_moral = ifelse(MoralityCondition == 102, quadratic_time_centered, 0)) %>%
  mutate(t3_moral = ifelse(MoralityCondition == 102, cubic_time_centered, 0)) %>%
  mutate(t1_immoral = ifelse(MoralityCondition == 103, linear_time_centered, 0)) %>%
  mutate(t2_immoral = ifelse(MoralityCondition == 103, quadratic_time_centered, 0)) %>%
  mutate(t3_immoral = ifelse(MoralityCondition == 103, cubic_time_centered, 0)) %>%
  filter((incCorr==1)&(incZygo==1))

#compute the % of data loss after baseline selection
perc_inc_data = 100-(dim(mor_long_inc)[1]/dim(mor_long)[1])*100
perc_inc_data
```

```
## [1] 2.578797
```

The baseline procedure has resulted in an exclusion of 2.5787966% of the data.

2. State Adjective segment

```r
# convert from wide to long format for time and select only corrugator
adj_long <- adj_data_raw %>%
  dplyr::select(!starts_with("StateAdjectiveZygoPr100ms")) %>%
  pivot_longer(
    cols = starts_with("StateAdjectiveCorrPr100ms"),
    names_to = "time",
    names_prefix = "StateAdjectiveCorrPr100ms_",
    names_transform = list(time = as.integer),
    values_to = "StateAdjective_response"
  )
str(adj_long)

# moral(102)* pos(104), neg(105) = 608, 710
# immoral(103)* pos(104), neg(105) = 712, 815

# compute relevant factors for the analysis
adj_long <- adj_long %>%
  mutate(MoralityValence = MoralityCondition*StateAdjectiveCondition-10000)
```

```r
#compute linear, quadratic and cubic time in order to compute time trends per condition
linear_time_centered <- (adj_long$time-8)/10
unique(linear_time_centered)
quadratic_time_centered <- linear_time_centered*linear_time_centered
cubic_time_centered <- linear_time_centered*quadratic_time_centered

#compute time trends per condition and filter only the included trials based on incCorr and incZygo
adj_long_inc <- adj_long %>%
  mutate(t1_moral_pos = ifelse(MoralityValence == 608, linear_time_centered, 0)) %>%
  mutate(t2_moral_pos = ifelse(MoralityValence == 608, quadratic_time_centered, 0)) %>%
  mutate(t3_moral_pos = ifelse(MoralityValence == 608, cubic_time_centered, 0)) %>%
  mutate(t1_immoral_pos = ifelse(MoralityValence == 712, linear_time_centered, 0)) %>%
  mutate(t2_immoral_pos = ifelse(MoralityValence == 712, quadratic_time_centered, 0)) %>%
  mutate(t3_immoral_pos = ifelse(MoralityValence == 712, cubic_time_centered, 0)) %>%
  mutate(t1_moral_neg = ifelse(MoralityValence == 710, linear_time_centered, 0)) %>%
  mutate(t2_moral_neg = ifelse(MoralityValence == 710, quadratic_time_centered, 0)) %>%
  mutate(t3_moral_neg = ifelse(MoralityValence == 710, cubic_time_centered, 0)) %>%
  mutate(t1_immoral_neg = ifelse(MoralityValence == 815, linear_time_centered, 0)) %>%
  mutate(t2_immoral_neg = ifelse(MoralityValence == 815, quadratic_time_centered, 0)) %>%
  mutate(t3_immoral_neg = ifelse(MoralityValence == 815, cubic_time_centered, 0)) %>%
  filter((incCorr==1)&(incZygo==1))
```

The baseline procedure has resulted in an exclusion of 2.5787966% of the data.

3. Affect Reason segment

```r
# convert from wide to long format for time and select only corrugator
reason_long <- reason_data_raw %>%
  dplyr::select(!starts_with("AffectReasonZygoPr100ms")) %>%
  pivot_longer(
    cols = starts_with("AffectReasonCorrPr100ms"),
    names_to = "time",
    names_prefix = "AffectReasonCorrPr100ms_",
    names_transform = list(time = as.integer),
    values_to = "AffectReason_response"
  )
str(reason_long)

# moral(102)* pos(104), neg(105) = 608, 710
# immoral(103)* pos(104), neg(105) = 712, 815

# compute relevant factors for the analysis
reason_long <- reason_long %>%
  mutate(MoralityValence = MoralityCondition*StateAdjectiveCondition-10000)

#compute linear, quadratic and cubic time in order to compute time trends per condition
linear_time_centered <- (reason_long$time-13)/10
unique(linear_time_centered)
quadratic_time_centered <- linear_time_centered*linear_time_centered
cubic_time_centered <- linear_time_centered*quadratic_time_centered
```

```r
#compute time trends per condition and filter only the included trials based on incCorr and incZygo
reason_long_inc <- reason_long %>%
  mutate(t1_moral_pos = ifelse(MoralityValence == 608, linear_time_centered, 0)) %>%
  mutate(t2_moral_pos = ifelse(MoralityValence == 608, quadratic_time_centered, 0)) %>%
  mutate(t3_moral_pos = ifelse(MoralityValence == 608, cubic_time_centered, 0)) %>%
  mutate(t1_immoral_pos = ifelse(MoralityValence == 712, linear_time_centered, 0)) %>%
  mutate(t2_immoral_pos = ifelse(MoralityValence == 712, quadratic_time_centered, 0)) %>%
  mutate(t3_immoral_pos = ifelse(MoralityValence == 712, cubic_time_centered, 0)) %>%
  mutate(t1_moral_neg = ifelse(MoralityValence == 710, linear_time_centered, 0)) %>%
  mutate(t2_moral_neg = ifelse(MoralityValence == 710, quadratic_time_centered, 0)) %>%
  mutate(t3_moral_neg = ifelse(MoralityValence == 710, cubic_time_centered, 0)) %>%
  mutate(t1_immoral_neg = ifelse(MoralityValence == 815, linear_time_centered, 0)) %>%
  mutate(t2_immoral_neg = ifelse(MoralityValence == 815, quadratic_time_centered, 0)) %>%
  mutate(t3_immoral_neg = ifelse(MoralityValence == 815, cubic_time_centered, 0)) %>%
  filter((incCorr==1)&(incZygo==1))
```

The baseline procedure has resulted in an exclusion of 2.5787966% of the data.